BM and Markov Processes

As we have seen before, BM has the following property:

$$W_t := B_{t+s} - B_s \qquad t \geq 0$$

and

$$B_t \qquad 0 \leq t \leq s$$

are independent (as processes)

In other words, since $B_{t+s} = W_t + B_s$,

$$P\left(B_{t+s} \in E \mid \mathcal{F}_s^B\right) = P\left(W_t + B_s \in E \mid \mathcal{F}_s^B\right)$$

we will make this rigorous.

$$= \underbrace{P\left(W_t + B_s \in E \mid B_s\right)}$$

$$= P\left(W_t + x \in E\right)\Big|_{x = B_s}$$

It is clear that we don't need in general to work with the natural filtration of BM but we can

Consider any admissible filtration, i.e., the following is true.

Theorem. Let $(B_t)_{t\geq 0}$ be a BM^d and $(\mathcal{F}_t)_{t\geq 0}$ some admissible filtration. For every $s>0$, the process $W_t := B_{t+s} - B_s$, $t\geq 0$, is also a BM^d and $(W_t)_{t\geq 0}$ is indep of $\mathcal{F}_s$, i.e., $\mathcal{F}_\infty^W := \sigma(W_t, t\geq 0)$ is indep of $\mathcal{F}_s$.

The point is that we can split any trajectories into two parts, which are <u>independent</u>:

take $s,t\geq 0$

$$B_{t+s} = \underbrace{B_{t+s} - B_s} + B_s$$

$$= W_t + B_s = W_t + y \Big|_{y=B_s}$$

independent

Note that $W_t + y$ is a BM started at y.

The notation

$$P^x(B_{t_1}\in A_1, \ldots B_{t_m}\in A_m) := P(x+B_{t_1}\in A_1, \ldots x+B_{t_m}\in A_m)$$

is very common for Markov processes.

The measure corresponding to $\mathbb{P}^x$ on the path space, i.e., the Wiener measure μ^x is supported on the continuous functions $w : [0,\infty) \longmapsto \mathbb{R}^d$ s.t. $w(0) = x$.
It follows also that

$$x \longmapsto \mathbb{E}^x u(B_t)$$

is measurable for every $u \in \underbrace{B_b(\mathbb{R}^d)}$

bounded Borel measurable functions $u : \mathbb{R}^d \longmapsto \mathbb{R}$.

this assertion merits a sketched proof:

Note that

$$\mathbb{E}^x u(B(t)) = \mathbb{E}\, u(x + B(t))$$

$$= \int u(x+y)\, \mathbb{P}(B(t) \in dy)$$

and this last representation is clearly measurable (Fubini).
Furthermore $x \longmapsto \mathbb{E}^x u(B_t)$ is also bounded, i.e.,

$$x \longmapsto \mathbb{E}^x u(B_t) \text{ is in } B_b(\mathbb{R}^d).$$

The last assertion can be generalized to any r.v. $Z \in \mathcal{F}_\infty^B$ bounded.

The most technical result will be crucial to make things rigorous.

Lemma. Let $X : \Omega \mapsto D$ s.t. $X^{-1} : \mathcal{D} \mapsto \mathcal{A}$

and $Y : \Omega \mapsto E$ s.t. $Y^{-1} : \mathcal{E} \mapsto \mathcal{A}$.

be two random variables. Suppose that $\mathcal{X}, \mathcal{Y} \subset \mathcal{A}$

are σ-Algebras s.t. $X^{-1} : \mathcal{D} \mapsto \mathcal{X}$ and

$Y^{-1} : \mathcal{E} \mapsto \mathcal{Y}$, — $\sigma(X) \subset \mathcal{X}$, $\sigma(Y) \subset \mathcal{Y}$

and further

$$\mathcal{X} \perp \mathcal{Y}.$$

Then

$$E[\phi(X,Y) \mid \mathcal{X}] = E\phi(x,Y)\big|_{x=X}$$

$$= E[\phi(X,Y) \mid X]$$

for every bounded functions ϕ s.t.

$$\phi^{-1} : \mathcal{B}(\mathbb{R}) \mapsto \mathcal{D} \times \mathcal{E}$$

Proof. Suppose first that $\phi(x,y) = \phi_1(x)\,\phi_2(y)$.

Then

$$E[\phi(X,Y) \mid \mathcal{X}] = E[\phi_1(X)\,\phi_2(Y) \mid \mathcal{X}]$$

$\phi_1(X) \in m\mathcal{X}$

$$= \phi_1(X)\, E[\phi_2(Y) \mid \mathcal{X}]$$

$$\mathcal{X} \perp \mathcal{Y} \quad Y \in m\mathcal{Y}$$

$$= \phi_1(X)\, E\, \phi_2(Y)$$

$$= \Big[E\, \phi(x, Y)\Big]_{x=X} \longrightarrow \Big[E\, \phi_1(x)\, \phi_2(Y)\Big]_{x=X}$$

By definition of conditional expectation this means that, for every $F \in \mathcal{X}$, one has

$$E\Big[1_F\, \phi(X,Y)\Big] = E\Big[1_F\, \big[E\phi(x,Y)\big]_{x=X}\Big]$$

whenever $\phi(x,y) = \phi_1(x)\, \phi_2(y)$. $\ast$

In other words, letting $\phi(x,y) = 1_A(x)\, 1_B(y)$ for $A \in \mathcal{D}$ and $B \in \mathcal{E}$, the equality $\ast$ reads

$$\int_F 1_{A\times B}(X,Y)\, dP = \int_F \Big[E\, 1_{A\times B}(x,Y)\Big]_{x=X} dP$$

for every $F \in \mathcal{X}$. Then the class $\underbrace{\{A\times B : A \in \mathcal{D}, B \in \mathcal{E}\}}_{=:\Sigma_1}$ and the mappings

$$\Sigma_1 \ni A\times B \longmapsto \int_F 1_{A\times B}(X,Y)\, dP$$

$$\Sigma_1 \ni A\times B \longmapsto \int_F \Big[E\, 1_{A\times B}(x,Y)\Big]_{x=X} dP$$

are countably additive maps (prove this). Then there the mappings can be extended to a (unique) measure on $\sigma(\Sigma')$. It follows that

$$\int_F \mathbb{1}_\sigma(X,Y)\,dP = \int_F \left[E\,\mathbb{1}_\sigma(x,Y)\right]_{x=X} dP \qquad \boxtimes$$

for every $\sigma \in \sigma(\Sigma')$ and $F \in \mathcal{X}$.

- indicator function
- SF^+ (by lim)
- (on Σ')$^+$ (by MON) - $B_b(D\times E)$ (by LIN)

By a classical standard machine argument we can extend the equality $\boxtimes$ to any function $\phi \in B_b(D\times E)$, to get

$$E\,\mathbb{1}_F\,\phi(X,Y) = E\left[\mathbb{1}_F\left[E\,\phi(x,Y)\right]_{x=X}\right]$$

which is equivalent to

$$E\left[\phi(X,Y)\mid\mathcal{X}\right] = \cancel{E\left[\left[E\,\phi(x,Y)\right]_{x=X}\mid\mathcal{X}\right]}$$

$$= \left[E\,\phi(x,Y)\right]_{x=X}$$

which is the first equality

Now take conditional expectation on both sides of the previous equality to get

$$\mathbb{E}\Big[\mathbb{E}\big[\phi(X,Y) \mid \mathcal{X}\big] \mid X\Big] = \mathbb{E}\Big[\big[\mathbb{E}\phi(x,Y)\big]_{x=X} \mid X\Big]$$

$\downarrow$ tower $\qquad\qquad\qquad\qquad \downarrow$ measurability

$$\mathbb{E}\big[\phi(X,Y) \mid X\big] = \big[\mathbb{E}\phi(x,Y)\big]_{x=X}$$

$$= \mathbb{E}\big[\phi(X,Y) \mid \mathcal{X}\big]$$

$\hookrightarrow$ by the 1^{st} equality.

We are ready to state the Markov property informally.

Theorem. Let $(B_t)_{t\geq 0}$ and $(\mathcal{F}_t)_{t\geq 0}$ be a BM and an admissible filtration then

$$\mathbb{E}\big[u(B_{t+s}) \mid \mathcal{F}_s\big] = \mathbb{E}^{B_s}\big[u(B_t)\big]$$

$$\Big(= \big[\mathbb{E}\, u(B_t + x)\big]_{x=B_s}\Big)$$

for any $u \in \mathcal{B}_b(\mathbb{R}^d)$.

Proof. Write $u(B_{t+s}) = u(\underset{=:X}{B_s} + \underbrace{B_{t+s} - B_s}_{=:Y})$.

Then $X \perp Y$ $X \in m\mathcal{F}_r$ $Y \perp \mathcal{F}_r$ and Thus

$$E[u(B_{t+r}) \mid \mathcal{F}_r] = E[u(X+Y) \mid \mathcal{F}_r]$$

use previous lemma with $\phi(x,y) = u(x+y)$

$$= \left[E\, u(x+Y)\right]_{x=X}$$

$$= \left[E\, u(x+B_{t+r}-B_r)\right]_{x=B_r}$$

stationary increments

$$= \left[E\, u(x+B_t)\right]_{x=B_r}$$

$$= E^{B_r} u(B_t).$$

Now we extend the previous result to any functional of Brownian trajectories. For this we need:

Lemma. Under the assumptions of the lemma above, suppose further that

$$\Psi : \Delta \times \Omega \mapsto \mathbb{R} \quad \text{is s.t.}$$

$$\Psi^{-1} : \mathcal{B}(\mathbb{R}) \mapsto \mathcal{Q} \otimes \mathcal{Y} \quad \text{and is bounded.}$$

Then $E[\Psi(X(\cdot), \cdot) \mid \mathcal{X}]$

$$= \left[E\, \Psi(x, \cdot)\right]_{x = X}$$

$$= E[\Psi(X(\cdot), \cdot) \mid X]$$

Proof. Very similar to the previous case since the dependence of $\Psi(X(\cdot), w)$ on w is s.t. it is "independent" among the two coordinates.

Use now this lemma to prove that

Theorem. Let $(B_t)_{t \geq 0}$ and $(\mathcal{F}_t)_{t \geq 0}$ be BM and adm. filtration. Then

$$E[\Psi(B_{\cdot + s}) \mid \mathcal{F}_s] = E^{B_s}[\Psi(B_\cdot)]$$

for all $\Psi : C[0,\infty) \mapsto \mathbb{R}$ s.t. $\Psi^{-1} : \mathcal{B} \mapsto \mathcal{B}(C)$ bounded.

Proof. Use previous lemma as follows.

$$\psi(B_{\cdot+r}) = \psi(B_{\cdot+r} - B_r + \underbrace{B_r}_{X}).$$

$$\mathcal{X} = \mathcal{F}_r \quad \mathcal{Y} = \mathcal{F}^{B_{t+r}-B_r}_{t\geq 0}$$

Then $\mathcal{X} \perp \mathcal{Y}$ and thus

$$E\left[\psi(B_{\cdot+r}) \mid \mathcal{F}_r\right] = E\left[\psi(B_{\cdot+r} - B_r + B_r) \mid \mathcal{F}_r\right].$$

$$= E\left[\psi(B_\cdot + x)\right]_{x=B_r}$$

$$= E^{B_r}\left[\psi(B_\cdot)\right]$$

Note that we used everywhere the abuses of notation $B_\cdot + x$, $B_{\cdot+r}$ to denote

$B_\cdot + x$ $\downarrow$ space translation of a function

$B_{\cdot+r}$ $\searrow$ time translation of a function

$\lrcorner$

Note that previous Theorem does not need to be applied to processes with stationary increments. In general, any $\mathcal{F}_t$-adapted random process is said to be a Markov process if

$$\mathbb{E}[u(X_{t+s}) \mid \mathcal{F}_s] = \mathbb{E}[u(X_{t+s}) \mid X_s]$$

for every $t \geq 0$ and $s \geq 0$ and $u \in \mathcal{B}_b(\mathbb{R}^d)$.

Denote

$$g_{u,s,t+s}(x) := \mathbb{E}[u(X_{t+s}) \mid X_s = x]$$

and define

$$\mathbb{E}^{s,x} u(X_{t+s}) := g_{u,s,t+s}(x).$$

If $g_{u,s,t+s}(x)$ depends only on $(t+s)-s=t$, i.e., the difference, then we call the process homogeneous and the Markov property is written as

$$\mathbb{E}[u(X_{t+s}) \mid \mathcal{F}_s] = \mathbb{E}^{X_s} u(X_t)$$

It is clear that, for homogeneous process this equality implies the more general

$$\mathbb{E}[u(X_{t+r}) \mid \mathcal{F}_r] = \mathbb{E}[u(X_{t+r}) \mid X_r].$$

Further for Markov processes one has

CAREFULL $\cdot + r \longrightarrow \omega + r$

$$\mathbb{E}[\Psi(\underbrace{X_{\cdot+r}}_{\omega+r}) \mid \mathcal{F}_r] = \mathbb{E}[\Psi(X_{\cdot+r}) \mid X_r]$$

(homogeneity) $$= \mathbb{E}^{X_r}[\Psi(\underset{\omega}{X_\cdot})]$$

for any bounded measurable functionals $\Psi : C[0,\infty) \longmapsto \mathbb{R}$ of the trajectories.

NB: Altough this last relationship looks more general it impossible to obtain it from the definition.

Remark. Recall the Markov property

$$\mathbb{E}[u(X_{t+r}) \mid \mathcal{F}_r] = \mathbb{E}^{X_r} u(X_t)$$

by specifying the dependence on "ω" or

$$\mathbb{E}\left[u(X_{t+r}) \mid \mathcal{F}_r\right](\omega)$$

$$= \int_\Omega u(X_t(\omega))\, dP^{X_r(\omega)}$$

$$= \int_{\mathbb{R}^d} u(y)\, P^{X_r(\omega)}(X_t \in dy) \quad (*)$$

Now take three different times $0 \leq r < t < u$ and $\phi\ \psi\ \chi : \mathbb{R}^d \mapsto \mathbb{R}$.

$$\mathbb{E}\left[\phi(B_r)\psi(B_t)\chi(B_u)\right] = \mathbb{E}\,\mathbb{E}\left[\phi(B_r)\psi(B_t)\chi(B_u) \mid \mathcal{F}_t\right)$$

$$= \mathbb{E}\left[\phi(B_r)\psi(B_t)\ \mathbb{E}\left[\chi(B_u) \mid \mathcal{F}_t\right]\right]$$

Markov prop.

$$= E\left[\phi(B_r)\psi(B_t)\, E^{B_t}\chi(B_{u-t})\right]$$

$$= \mathbb{E}\, E\left[\phi(B_r)\psi(B_t)\, E^{B_t}\chi(B_{u-t}) \mid \mathcal{F}_r\right]$$

$$= \mathbb{E}\left[\phi(B_r)\ E\left[\psi(B_t)\, E^{B_t}\chi(B_{u-t}) \mid \mathcal{F}_r\right]\right]$$

Markov property

$$= E\left[\phi(B_r)\quad E^{B_r}\psi(B_{t-r})\, E^{B_{t-r}}\chi(B_{u-t})\right]$$

use $\circledast$ repeatedly

$$= \left\{ \int \phi(x) \left[\int \psi(y) \left(\int \chi(z) \, P^{y}(B_{u-t} \in dz) \right) P^{x}(B_{t-s} \in dy) \right] P(B_s \in dx) \right\}$$

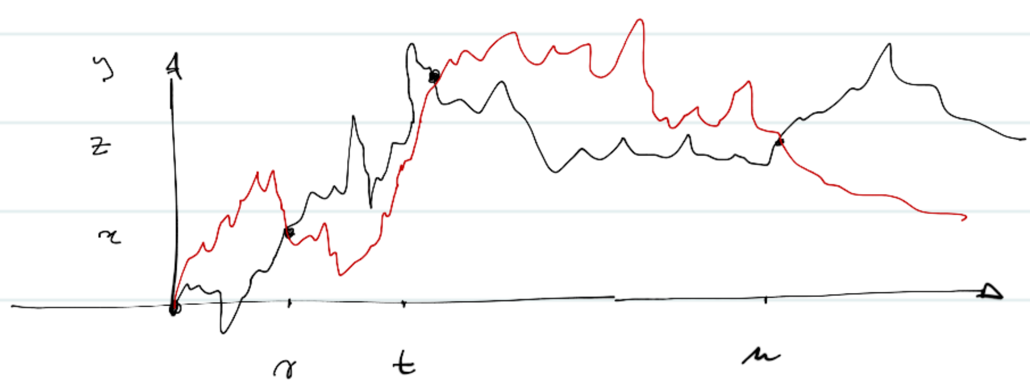
The above equality reads

$$\iiint \phi(x) \psi(y) \chi(z) \, P(B_s \in dx, B_t \in dy, B_u \in dz)$$

$$= \iiint \phi(x) \psi(y) \chi(z) \; P^{y}(B_{u-t} \in dz) \, P^{x}(B_{t-s} \in dy) \, P(B_s \in dx)$$

and this essentially means that the joint distribution (B_s, B_t, B_u) is

$$P(B_s \in dx, B_t \in dy, B_u \in dz) = P^{y}(B_{u-t} \in dz) \, P^{x}(B_{t-s} \in dy) \, P(B_s \in dx)$$



Prop. of these trajectories:

$$0 \longrightarrow x$$

$$x \xrightarrow[t-s]{} y$$

$$y \xrightarrow[u-t]{} z$$

With this in mind it is possible to prove:

Theorem. Let (X_t) be a (homogeneous) Markov process, let $t_0 = 0 < t_1 < \dots < t_n$. Then

$$P(X_{t_1} \in dx_1, \dots, X_{t_n} \in dx_n) = \prod_{j=1}^{n} P^{x_{j-1}}(X_{t_j - t_{j-1}} \in dx_j).$$

fin-dim marginal!